Privacy-First Personal Finance Agent: ReceiptIQ
Individual Project Report
Role: Backend Developer
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Introduction

This report documents my contribution as the Backend Developer for the ReceiptIQ project. My work focused on backend system design, ensuring efficient data management, and maintaining system reliability.

Role Summary

I was responsible for implementing backend logic including receipt storage, transaction handling, data flow management, and export functionality. The backend ensures smooth coordination across modules.

Backend System Design

The backend follows a modular pipeline architecture connecting OCR, preprocessing, ML, and frontend layers. It ensures structured data flow and system stability.

Core Responsibilities

• Receipt storage management
• Transaction processing
• Data flow between modules
• Export system implementation

Local Storage Design

A local-first design improves performance, enables offline access, ensures privacy, and removes server dependency.

Duplicate Detection

A hash-based detection system prevents duplicate receipt entries by comparing unique identifiers.

Additional Features

Includes CSV export, lightweight record management, and fast expense history retrieval.

Key Learnings

1. Backend structure defines system efficiency.
2. Privacy-first design improves trust.
3. Data validation is essential.
4. Simplicity leads to better performance.

Conclusion

The backend system ensures reliable, efficient, and secure operation of ReceiptIQ through optimized architecture.

Individual Contribution Statement

This report describes my independent backend development contribution including system design and implementation.